

MANISH KUMAR

Boston, MA +1 669-336-2564 manishkr2106@gmail.com [linkedin.com/in/manish2106](https://www.linkedin.com/in/manish2106) [manishkr2106.github.io](https://github.com/manishkr2106)

PROFESSIONAL SUMMARY

Data-driven strategy and sustainability professional pairing a STEM MBA in Business Analytics & Machine Learning with 7+ years of civil and water-resources engineering on World Bank-funded infrastructure. Translate ambiguous business problems into measurable outcomes by combining supply-chain optimization, predictive modeling, and ESG fluency with hands-on engineering execution across consulting, analytics, and clean-energy contexts.

EDUCATION

BABSON COLLEGE, F.W. OLIN GRADUATE SCHOOL OF BUSINESS, Wellesley, MA

September 2024 – May 2026

STEM MBA Candidate | Business Analytics & Machine Learning / Entrepreneurship / Supply Chain

- **Supply-Chain Gating Pipeline (Apache Airflow):** Built and validated an Airflow DAG orchestrating a daily extract–transform–gating–reporting pipeline for multi-tier supply visibility, computing days-of-supply coverage across parts and suppliers and producing a risk-ranked report of gating parts whose coverage fails to span lead time — with retry handling and scheduled execution for automated, predictive stock alerting.
- **Zara Reverse-Logistics Optimization (MILP):** Designed a mixed-integer linear-programming reverse-logistics network in Python (PuLP) integrating EOQ, Newsvendor, and transportation optimization to minimize cost while maximizing circularity across a global fashion supply chain, with blockchain-traceability and AI-enhancement proposals.
- **Credit Default Prediction (ML):** Built a six-model credit-risk pipeline in R on 150K borrower records (logistic regression, CART, neural network, stacking); a class-weighted neural network reached 0.86 AUC and lifted sensitivity from 4% to 77.8% by tuning the probability threshold to the business cost of a missed default rather than raw accuracy.
- **IBM Watsonx Library Assistant:** Designed and deployed an AI research assistant connecting Babson's Horn Library Primo REST API to IBM Watsonx Orchestrate via a Node.js / MCP server on Railway.app — owning data modeling, production workflow integrity, and institutional IT handover documentation.
- **RentSmart Boston:** Developed an interactive Tableau dashboard on 10,000+ rental data points backed by a full Python ETL pipeline (extraction, validation, cleaning), delivering neighborhood-level affordability insights as stakeholder-ready recommendations.
- **Relational Database & BI Reporting:** Designed a 9-table 3NF SQLite database with a Lucidchart ERD, validated imports in DBeaver, and wrote 8 BI SQL queries (CTEs, window functions, multi-table joins) covering budget-vs-spend variance, workload, portfolio status, and client profitability.
- **Peacock Streaming Strategy:** Applied the Value Net and PARTS frameworks to size the competitive landscape and analyze growth vectors in streaming, synthesizing findings into an executive-ready strategy memo.
- **Predictive Modeling & NLP:** Built ML classification pipelines (CART, Random Forest, GBM, stacking) in R and NLP sentiment/topic modeling (TextBlob, BERTopic, IBM Watsonx.ai) over large text datasets.

MANAV RACHNA INTERNATIONAL UNIVERSITY, Faridabad, Haryana, India

July 2013 – November 2017

Bachelor of Technology | Civil Engineering

EXPERIENCE

CHOPVALUE BOSTON, Boston, MA

January 2025 – May 2025

A circular-economy B-Corp that transforms used chopsticks into carbon-negative, sustainable products for modern living.

Consultant (Babson Consulting Experience)

- Led an end-to-end consulting engagement from intake through closeout — framed the problem, built a fact base from a 50+ firm competitive-intelligence dataset, and sized the market for recycled wood panels in LEED-certified commercial segments, delivering executive-ready go-to-market recommendations.
- Built a regression-based pricing and price-elasticity model across three scenarios, synthesizing 20+ stakeholder interviews to quantify demand sensitivity and project 2x revenue potential that informed the client's pricing strategy.
- Developed a three-tier customer segmentation from willingness-to-pay data, uncovering a 40% WTP premium among mission-aligned buyers that reshaped the client's go-to-market priorities.
- Maintained data-validation checkpoints across all reporting workflows and translated the analysis into a 30-slide executive presentation and 12-month roadmap adopted by client leadership.

NAWAL SINGH CONSTRUCTIONS PRIVATE LIMITED, Patna, Bihar, India

April 2021 – July 2024

*A privately held construction firm focused on sustainable infrastructure, prioritizing community impact.***Project Manager — Water Resources Department**

- Owned scope and execution of a 10+ project, \$18M infrastructure portfolio (including World Bank-funded programs), directing a 55-member cross-functional team from intake through closeout.
- Applied critical-path scheduling and resource optimization to cut project delays 15% and deliver ~10% cost savings across the portfolio.
- Designed and governed a QA/QC measurement-and-billing system — establishing audit logic and change controls — that reduced defect rates 25% and saved ~\$500K annually in rework and penalties.
- Led user acceptance testing (UAT) against live workflows (measurement entries, RA-bill preparation, drawing revision control), triaging defects by billing and compliance risk before deployment to safeguard government billing and audit accuracy.
- Consolidated reporting from 5+ field teams into KPI dashboards in Microsoft Dynamics 365 and Power BI, cutting reporting lag 40% and giving leadership and World Bank reviewers an audit-ready, real-time portfolio view.
- Managed procurement and vendor coordination — requisitions, ordering, shipment tracking — and sustained critical-path operations through COVID-19 border shutdowns by dynamically reallocating inventory across competing sites, preventing complete work stoppage for 14 consecutive days.
- Performed demand-versus-available-supply analysis and built inventory allocation models that balanced limited stock across competing sites, and ran supply simulations on shortages, lead times, and depletion timelines to flag gating risks and drive proactive contingency planning.
- Authored ESG impact reports aligned to GHG protocols and SDG indicators for World Bank compliance review, integrating energy, water, and emissions data into decarbonization-aligned dashboards.

WHALE BLUE HITECH PRIVATE LIMITED, India

January 2019 – March 2021

*Water-resources and flood-control engineering firm serving government and community infrastructure programs.***Assistant Engineer**

- Led watershed-modeling projects that improved hydrological efficiency and water quality for 10,000+ residents.
- Supervised inspection and maintenance of flood-control structures, increasing operational uptime and extending embankment service life.
- Mentored 25 junior engineers through on-site workshops, raising technical capability and project throughput.
- Conducted terrain and hydrological assessments using HEC-RAS, GIS, SLIDE, and GEO-SLOPE for data-driven infrastructure design, and coordinated with stakeholders to ensure compliance with regional water-resource standards.

Site Engineer (October 2017 – December 2018)

- Managed day-to-day field operations across civil infrastructure sites, ensuring timelines, quality, and safety compliance.
- Supported construction of irrigation and flood-control systems including canal lining, embankment reinforcement, and culvert installation.
- Conducted soil testing, topographic surveys, and slope-stability checks (AutoCAD, STAAD.Pro, SLIDE) to inform design changes and construction planning, and coordinated vendors to keep material supply aligned with specifications.

CENTRAL WATER COMMISSION, India

June 2016 – December 2016

*National water-resources authority; Directorate CMDD (NWS).***Engineer Intern**

- Gained exposure to major national water-infrastructure projects (Garada Embankment Dam, Polavaram Dam), studying load combinations, stress analysis, and stability against sliding, overturning, and uplift per IS codes.
- Assisted with AAA and Class-A live-load analysis per IRC standards and applied SLIDE / GEO-SLOPE to slope-stability analysis of dam embankments and bridge approach zones.

TECHNICAL SKILLS & CERTIFICATIONS

- **Languages & Query:** Python (pandas, NumPy), R, SQL (large-scale querying & reporting), C, C++, Java
- **Data, Pipelines & Automation:** ETL / data-pipeline development, Apache Airflow orchestration, automated reporting, data-validation workflows, data modeling (3NF / relational schema design), containerized deployment (Railway.app, MCP server architecture), low-code / REST API integration
- **Analytics & Modeling:** Statistical modeling, regression & forecasting (ARIMA, AIC/BIC), price-elasticity modeling, A/B testing, customer segmentation, market sizing, scenario analysis, Monte Carlo simulation

- **Machine Learning:** Supervised & unsupervised learning, Random Forest, CART, GBM, kNN, Naïve Bayes, neural networks (backpropagation), stacking ensembles, NLP (sentiment / topic modeling), ROC / k-means / association rules
- **Supply Chain & Optimization:** Tactical procurement & PO coordination, demand–supply planning, inventory optimization (EOQ, Newsvendor, safety-stock, reorder-point, joint replenishment, constrained optimization), supplier-disruption & gating analysis, MILP network design, facility location, transportation planning, Gurobi, PuLP, NetworkX
- **Strategy & Consulting:** Competitive intelligence, business-case development, financial modeling, systems thinking, game theory (Value Net / PARTS), stakeholder facilitation, executive-ready synthesis
- **Business Intelligence & Visualization:** KPI frameworks (leading / lagging indicators), Power BI, Tableau, advanced Excel / Google Sheets (Pivot Tables, XLOOKUP / VLOOKUP), Matplotlib, interactive dashboards, Jupyter Notebook, HTML / CSS / JavaScript web platforms
- **Business Analysis:** Requirements elicitation & analysis, functional specifications, business-to-IT translation, user acceptance testing (UAT), change management, process design & governance, audit & compliance, internal controls
- **Enterprise Systems & Process:** Microsoft Dynamics 365 (D365), SmartSheets, ERP / PLM data structures, master-data governance (SKU / BOM), procurement & purchase requisitions, billing / invoicing coordination, order fulfillment & shipment tracking, SharePoint
- **AI & Emerging Tech:** IBM Watsonx.ai, IBM Watsonx Orchestrate (AI agent development), MCP server integration (Node.js), REST APIs, Microsoft Copilot, Databricks, AI-enabled solutions design
- **Sustainability & ESG:** GHG protocols (Scope 1/2/3), ESG metrics, GRI, CDP, EcoVadis, SDG-aligned reporting, decarbonization / net-zero, Distributed Energy Resources (DER), energy / water / emissions analysis
- **Engineering Tools:** AutoCAD, STAAD.Pro, HEC-RAS, HEC-HMS, SWMM (EPA), SLIDE, GEO-SLOPE, GIS, Vensim (systems modeling)
- **Tools & Process:** Jira (Agile workflows), Confluence, Git / GitHub, PowerPoint, Word, Agile
- **Certifications:** Complete Project Management Course; Complete SQL Bootcamp 2022; Machine Learning A-Z (Python & R); Mastering Data Visualization; Introduction to Management Consulting (Emory University); Data Analytics (Google); Generative AI